



FEMIEL JUBIL M. TAMBUNAN

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Sumatera Utara, Indonesia

Fresh Electrical Engineering graduate from IT Del focused on control systems, automation, electrical systems, and critical infrastructure

Education Level

Institut Teknologi Del - Laguboti, Toba, Sumatera Utara, Indonesia
Bachelor of Electrical Engineering – Control Systems, 3.34/4.00

Jul 2022 - Aug 2026

Work Experiences

PT Wilmar Nabati Indonesia - Dumai, Riau, Indonesia

Jun 2025 - Aug 2025

Electrical Engineering Intern – Project Management Department, Electrical Division

- Conducted a study on the control system and operational testing of an electric motor used in a 1.5 MW Thermal Oil Heater system.
- Observed and supported Quality Control (QC) checks on Motor Control Centre (MCC) panels and electrical field activities.
- Participated in continuity testing of Electric Heat Tracing (EHT) installations and observed electric motor termination activities.
- Gained practical exposure to MCC panel finalization, industrial electrical installation, motor control systems, and field engineering workflows.
- Applied and observed occupational safety procedures while working around industrial electrical equipment and project activities.

Institut Teknologi Del - Laguboti, Toba, Sumatera Utara, Indonesia

Sep 2025 - Jul 2026

Student Researcher – Internal Research Grant

- Contributed as a student researcher in a 2026 IT Del Internal Research Grant project focused on the development of a modular Unmanned Surface Vehicle (USV) for long-range missions.
- Supported the integration and evaluation of navigation, telemetry communication, electrical power, and operational monitoring subsystems within the USV platform.
- Participated in system development and testing involving Pixhawk 6C, Raspberry Pi, Mission Planner, telemetry communication, and onboard electrical systems.
- Contributed to research activities under the project “Design and Performance Evaluation of a Modular Unmanned Surface Vehicle Integrating Navigation, Communication, and Operational Stability for Long-Range Missions.”

Institut Teknologi Del - Laguboti, Toba, Sumatera Utara, Indonesia

Sep 2025 - Dec 2025

Electronics II Laboratory Teaching Assistant

- Assisted students during Electronics II laboratory sessions and supported the implementation of practical experiments.
- Guided students in understanding electronic circuit concepts, measurement procedures, and laboratory equipment usage.
- Supported troubleshooting of electronic circuits during practical sessions and helped ensure laboratory activities were conducted safely.
- Assisted instructors in preparing and coordinating laboratory activities

Organisational Experience

Electrical Engineering Student Association (HME) – Institut Teknologi Del - Supported communication and public relations activities within the Electrical Engineering Student Association.

Jan 2024 - Dec 2024

Public Relations Officer

- Coordinated external communication with student organizations and partner institutions.
- Supported the development and maintenance of external partnerships for HME activities.
- Disseminated organizational information and coordinated communication between HME and external parties.

Line Follower Competition – HME Institut Teknologi Del - Toba, Sumatera Utara, Indonesia

Oct 2023 - Nov 2023

Committee Member

- Supported the preparation and implementation of a Line Follower Competition for senior high school students in Toba Regency.
- Assisted committee coordination and event preparation activities.
- Contributed to the execution of competition activities related to robotics and line-following technology.

Arduino Workshop – HME Institut Teknologi Del - Toba & North Tapanuli, North Sumatra, Indonesia

May 2024 - Jun 2024

Arduino Uno Workshop Instructor

- Introduced basic Arduino Uno and microcontroller concepts to senior high school students in Toba and North Tapanuli.
- Guided students through hands-on exercises using Arduino Uno and electronic components.
- Assisted students in understanding basic sensor integration and embedded system concepts.

Student Matriculation Program – Institut Teknologi Del - Laguboti, Toba, Sumatera Utara, Indonesia

Aug 2023

Academic Mentor

A program designed to support new students in adapting to the academic system and university environment.

- Provided academic guidance to new students on basic study strategies and university learning practices.
- Introduced the university's Learning Management System (LMS) and basic academic ethics.
- Supported new students in adapting to the academic and social environment of Institut Teknologi Del.

Department Member – Order and Discipline - Laguboti, Toba, Sumatera Utara, Indonesia

Sep 2023 - Sep 2024

Department Member

Student organization supporting student development, campus activities, and organizational coordination.

- Supported departmental activities related to student order, discipline, and campus regulations.
- Assisted in communicating applicable student rules and organizational information.
- Collaborated with other department members in supporting student activities and maintaining an orderly campus environment.

Skills, Achievements & Other Experience

- **Supervisory System for Unmanned Surface Vehicle (USV)** (2026): Designed a supervisory mechanism to maintain USV mission continuity and data availability during telemetry communication loss. Developed a monitoring dashboard for USV position and mission status with data storage during communication loss and retransmission after connectivity was restored. Integrated Pixhawk 6C, Raspberry Pi, telemetry radio, and Mission Planner for autonomous USV operation and monitoring. Contributed to the integration of a solar-powered electrical system and the development of a modular catamaran USV.
- **Digital Image Processing-Based Attendance System** (2026): Developed an attendance system as a final project in Digital Image Processing by applying image processing techniques to support automated attendance identification and practical implementation of computer vision concepts.
- **Rice Field Irrigation Automation System** (2025): Designed a monitoring and control system for rice field irrigation using the ATmega8535 microcontroller, applying embedded systems and automation concepts to support automated irrigation operation.
- **16-bit Single-Cycle MIPS Processor in VHDL** (2024): Designed and implemented a 16-bit single-cycle MIPS processor in VHDL using Intel/Altera FPGA tools, integrating the datapath, control unit, register file, ALU, memory, and branch/jump logic, with simulation and verification through an automated testbench.
- **IoT-Based In-Vehicle Temperature Monitoring** (2024): Developed an IoT-based system for real-time in-vehicle temperature monitoring as a safety-oriented application, applying sensor integration, data monitoring, and Internet of Things concepts.
- **Oscillator-Based Bell System** (2023): Designed and implemented an oscillator-based bell system as an analog electronics project, applying oscillator circuit principles to develop a practical electronic signaling system.
- **Arduino Uno Sensor and Servo Control Project** (2022): Developed a first-year group project using Arduino Uno, a raindrop sensor module, and a servo motor as an introduction to basic electrical engineering concepts, including microcontroller programming, sensor interfacing, electronic circuits, and actuator control.
- **Hard Skills**: Control Systems, Electrical Systems, Electric Motors, Motor Control Centre (MCC) Fundamentals, PLC Fundamentals, Arduino Uno, ATmega8535, Raspberry Pi, Pixhawk 6C, IoT, Sensor Integration, Telemetry, VHDL, FPGA, Digital Image Processing, Embedded Systems.
- **Operating Systems & Computing**: Ubuntu Linux, Raspberry Pi Linux Environment, Oracle VirtualBox.
- **Soft Skills**: Teamwork, Technical Communication, Problem Solving, Adaptability, Collaboration, Continuous Learning, Safety Awareness.
- **Interest**: Data Center Critical Facilities, Electrical Power Infrastructure, Industrial Automation, Control Systems, Electrical Design & Power System Analysis (AutoCAD Electrical, ETAP, Revit MEP), SCADA/BMS/EPMS, IoT, Autonomous Systems, Renewable Energy, and Machine Learning for Engineering Applications.
- **Engineering Software & Tools**: MATLAB, Proteus, NI Multisim, Mission Planner, Maxsurf, Intel/Altera FPGA Tools, GitHub.